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Challenge Name: A Sealed Record

Category: Digital Forensics

Approach Taken:

1.I downloaded and extracted the Artifact-BC34.rar archive and tried opening the PNG image. The image failed to open, indicating possible corruption in its file header.

2.I inspected the PNG header and checked the IHDR chunk, which stores the image dimensions and other properties. The CRC checksum did not match the current header data.

3. I analyzed the decompressed image data using Python to determine the correct image height. The calculation indicated that the original height was 1728 pixels.

4. I corrected the height value in the PNG header and verified the CRC checksum. The checksum matched the stored value, confirming that the header was repaired correctly.

5. I opened the repaired image and examined its contents to recover the record.

Challenge Name: 3000 Files Later

Category: Forensics

Approach Taken:

1. I extracted the get_a_life.zip archive and inspected its directory structure. It contained thousands of text files with randomly generated filenames and Base64-encoded content.

2. I decoded sample files and noticed that many contained convincing-looking flags with misleading keywords. Since the challenge warned against searching directly, I focused on the archive metadata and embedded clues.

3. I examined the ZIP archive comment, which referred to the beginning of an era and signing. I used this clue to investigate file timestamps and the information contained in the records.

4. I compared the candidate records using their timestamps and signature-related clues to narrow down the search. I then checked the relevant hash or signature information rather than trusting the flag text alone.

5. Finally, I verified the selected record against the identified clue and checked that its decoded content followed the expected flag format.

Challenge Name: Transmission

Category: Misc

Approach Taken:

1. I downloaded the provided transmission_v3.wav file and checked its basic properties, including its duration, sample rate, and audio format.

2. I listened to the recording in separate sections to check whether it contained speech, recognizable sounds, or an unusual transmission pattern.

3. I examined the audio spectrogram to identify repeated tones, frequency patterns, and unusual signal activity that might not be obvious during normal playback.

4. I investigated the suspicious signal sections to determine whether the audio contained encoded information. I also considered the possibility of hidden data that could be extracted using audio-analysis tools.

5. I used the challenge description as a clue to connect the audio evidence with the accompanying notes, since both sources were intended to contribute to understanding the transmission.

Challenge Name: Afterlife

Category: PWN

Approach Taken:

1. I downloaded the challenge.zip archive and extracted its contents. I inspected the provided files to understand how the archival service stored and managed records.

2. I examined the source code and configuration files to understand how records were created, deleted, and recovered by the service.

3. I traced the deletion process to check how the service updated record status and maintained internal bookkeeping information.

4. I analyzed the recovery mechanism to identify inconsistencies between deleted records and their stored metadata. I investigated how these inconsistencies could allow a supposedly deleted record to be accessed again.

5. I used the identified weakness to determine how the deleted record could be recovered and examined the recovered data to locate the flag.

Challenge Name: Afterimage

Category: Steganography

Approach Taken:

1. I downloaded the afterimage.png file and opened it to inspect the photograph. The image showed a large fire in a city at night, which suggested that additional information might be hidden in the image.

2. I checked the image properties, including its dimensions, color channels, and metadata, to look for unusual values or hidden clues.

3. I analyzed the image using steganography techniques to check whether any information was concealed in the pixel data or color channels.

4. I examined the image for hidden messages and checked whether the visual details or image data revealed information that was not immediately visible.

5. I used the relevant clues from the image to investigate the hidden content and determine how the information left behind could be recovered.

Challenge Name: The Margin Machine

Category: Cryptography

Approach Taken:

1. I examined the machine_card.txt and operator_log.txt files to understand the encryption machine and its working rule. The logbook contained three sample input/output pairs that could be used to analyze the cipher.

2. I checked the machine card, which described letter values from A=0 to Z=25 and a repeating key, LANTERN. It also stated that non-letter characters remain unchanged.

3. I noticed the handwritten margin note saying that the wheel now turns backward. This suggested that the original instruction to add key values was outdated and that subtraction modulo 26 should be used instead.

4. I compared the sample pairs with the stated key and tested the backward-shifting rule to understand how each letter was transformed. I also checked how the repeating key aligned with the message characters.

5. I applied the reconstructed cipher rule to the encoded message and used the sample data to check whether the decrypted text followed the expected format.

Challenge Name: The Engineer’s Broadcast

Category: Cryptography

Approach Taken:

1. I examined the operator_note.txt and recipients files to understand how the message was encrypted and distributed to the three recipients.

2. I checked the RSA public moduli, public exponents, and ciphertexts to determine whether the three recipients received the same encrypted payload.

3. I noticed that the operator's note confirmed that the same plaintext was encrypted using textbook RSA without randomized padding. This suggested that the broadcast attack could be applicable.

4. I investigated the relationship between the ciphertexts and their RSA moduli. Since the public exponent was small and the same plaintext was used, I checked whether the Chinese Remainder Theorem could reconstruct the plaintext raised to the public exponent.

5. I recovered the original message by combining the ciphertexts using the appropriate mathematical method and checked the decoded result for the expected flag format.

Challenge Name: Archive_042

Category: Cryptography

Approach Taken:

1. I downloaded instance.json and examined the modulus, base, nonce, and the two encrypted records. I used the challenge description to understand how the archive entries were generated.

2. I checked how each record's element was related to the public base and modulus. This helped me understand how the clerk's secret exponent was represented in the archive.

3. I investigated the structure of the modulus to determine whether its mathematical properties could weaken the secret-exponent mechanism. I also considered how the record elements could reveal information about the secret.

4. I examined the SHA-256 counter-mode keystream construction and the provided nonce to understand how the encrypted record data could be decrypted once the required secret value was recovered.

5. I analyzed both records separately and used the recovered cryptographic information to identify the relevant plaintext and determine which record contained the expected flag.

Challenge Name: The Tower in the Salt

Category: OSINT

Approach Taken:

1. I downloaded and read the desc.txt file to understand the historical clues about Manek, Govind, and Uncle Daya's submerged tower.

2. I identified the key geographical details, including the earthquake near Bhuj, the river that changed its course, and the former customs post on the route towards Sindh.

3. I researched the historical earthquake and its impact on the region to narrow down the possible location of the submerged fort and the lake formed after the disaster.

4. I used the clues about the Rao's customs post, the eastern riverbank, and the salt flats to compare possible locations with historical maps and satellite imagery.

5. I cross-checked the geographical evidence to identify the likely location of Uncle Daya's tower and used the map interface to mark the location as required by the challenge.

Challenge Name: Quiet Sum

Category: Cryptography

Approach Taken:

1. I downloaded instance.json and examined the RSA public key, ciphertext, and five telemetry readings provided in the file.

2. I compared the telemetry values to identify repeated readings and small differences between them. These variations suggested that the debug output might reveal information about an internal register.

3. I analyzed the relationship between the telemetry readings and the RSA key-generation process to determine whether the register values could leak useful information about the private key.

4. I investigated possible side-channel techniques and mathematical properties that could help recover the RSA private key from the leaked information, without requiring direct access to the hardware security module.

5. I used the recovered cryptographic information to decrypt the ciphertext and checked the resulting plaintext for the expected flag format.

Challenge Name: Echoes in the Archive

Category: Cryptography

Approach Taken:

1. I downloaded capture.csv and incident_report.txt and inspected the encrypted packet records. I noted the nonce, plaintext length, and ciphertext fields for each packet.

2. I read the incident report and used the known plaintext for pkt_00 as a starting point. This helped me investigate how the plaintext was transformed into ciphertext.

3. I compared packets that shared the same nonce to check whether the encryption system had reused the same keystream. Reusing a nonce in certain stream-cipher designs can expose relationships between encrypted messages.

4. I used the known plaintext and its ciphertext to recover the corresponding keystream bytes, then applied the recovered bytes to other records with the same nonce. I also considered the report's warning that some records were decoys created after the patch.

5. I examined the recovered plaintexts to find the record containing the sentence mentioned in the incident report. I then checked the relevant record's contents and verified the complete flag format.